Optics and Lasers in Engineering 201 (2026) 109627

Contents lists available at ScienceDirect

Optics and Lasers in Engineering

journal homepage: www.elsevier.com/locate/optlaseng

Optimization-based structured light profilometry for surface reconstruction of moving objects

Stefan Hager [a,b,*], Georg Schitter [a], Ernst Csencsics [a,b]

[a] Mechatronics and Power Electronics Institute TU Wien, Vienna, Austria
[b] Christian Doppler Laboratory for Precision Measurements in Motion, Vienna, Austria

ARTICLE INFO

Keywords:
Structured light profilometry
Surface reconstruction
Optical measurement

ABSTRACT

This paper presents a novel approach for accurate surface reconstruction of uniformly moving rigid samples from multi-shot structured light profilometry (SLP). Conventional triangulation-based multi-shot SLP cannot handle moving objects as motion during the acquisition phase causes errors in the assignment of pixel correspondence between cameras and projector. The proposed method utilizes an optimization-based, iterative strategy that considers the movement of the sample between consecutive captures to reconstruct the surface geometry. The optimization process is designed to minimize an objective function which evaluates the pixel value similarities between reprojected surface points in both the camera and projector images. The surface is reconstructed by iteratively refining the minimum search in the objective function using image filtering methods and gradient-based solvers. The method enables robust surface reconstruction of moving samples with a high measurement accuracy of 17 µm for various directions and extents of motion, which can compete with static measurement results from conventional, high-accuracy SLP methods while outperforming them by a factor of more than 20 for large sample displacements, making multi-shot SLP accessible for industrial in-line measurement applications.

1. Introduction

Three-dimensional (3D) surface measurements have become essential across a wide range of scientific and industrial applications, enabling precise geometric analysis and digital representation of physical objects. From industrial quality control and robot vision to dental scanning and cultural heritage preservation, various 3D scanning technologies have been developed that allow precise inspection and replication of object surfaces [1–5]. In industrial environments, 3D surface measurements ensure that products meet stringent tolerances, contributing to better quality control and reduced manufacturing errors [6].

A growing trend in modern manufacturing is the integration of in-line metrology into production lines, enabling continuous monitoring of both, process and product [7]. This capability is increasingly important in the context of automated manufacturing processes that require high levels of precision and efficiency. The ability to conduct 3D in-line measurements is particularly valuable, as it enables real-time inspection and validation of complex geometries, ensuring that products are manufactured to exact specifications while minimizing material waste and downtime due to errors or defects. A particular challenge, however, is the utilization of precision measurement techniques in the harsh environment of manufacturing lines, where heavy machinery and actuators, such as robotic arms and conveyor belts, introduce severe vibrations and sample motion [8].

Among various 3D measurement techniques, structured light profilometry (SLP) stands out due to its high precision, speed, and flexibility [9,10]. SLP operates by projecting known patterns onto the surface of an object and capturing the perspective deformations in these patterns using one or more cameras. These deformations caused by the object's surface topography are then analyzed to reconstruct the 3D shape. Depending on the number of projected patterns, SLP techniques are divided into single-shot and multi-shot techniques [11,12]. Single-shot SLP employing spatially-coded patterns, such as pseudo-random patterns, a De Bruijn-sequence or color-encoded stripes, aims to reconstruct the surface topology from a single capture, enabling reconstruction of dynamic scenes at high framerates of more than 30 fps [13–16]. However, the limited amount of information in the single capture constrains the spatial resolution to the size of a few projector pixels, obtaining an accuracy of about 0.4 mm [17,18]. For higher accuracy, as demanded in industrial applications, multi-shot SLP employs a sequence of projection patterns such as phase-shifted sinusoidal (PSS) images or binary-reflected Gray-coded (GC) patterns, which represent a temporally-coded set [19–21]. Multi-shot methods assume static conditions during the acquisition process in order to decode the temporally-coded pattern sequence

* Corresponding author.
E-mail address: stefan.hager@tuwien.ac.at (S. Hager).

https://doi.org/10.1016/j.optlaseng.2026.109627
Received 20 October 2025; Received in revised form 8 January 2026; Accepted 12 January 2026
Available online 28 January 2026

properly, leading to erroneous measurements of moving samples. The limitation to static conditions poses a challenge for applications in which objects are continuously moving, such as in production lines, where high throughput is essential and halting objects is inefficient [22,23]. While well-characterized uniform motion, induced by conveyor belts or robotic arms, can be beneficial for scanning measurement systems such as laser line triangulation [24], it decreases the quality of multi-shot SLP measurement results significantly.

To address this challenge, various approaches have been proposed [25]. The intuitive approach to make multi-shot SLP available for moving samples is to increase the acquisition rate of the system [26] or reduce the number of projected images [27], which enables the surface reconstruction of moving samples while decreasing the precision by more than a factor of three [28]. Additionally, motion effects due to small sample displacements can be detected and corrected by locally analyzing the resulting ripple on the reconstructed surface [29–31]. In [32], motion-induced errors are reduced using the information about the sample displacement for approximate correction of the phase retrieved from three projected PSS images. Interlaced patterns for tracking the displacement of projector pixels in the camera images were qualitatively evaluated in [33], leading to an increased number of projections and a semi-accurate result due to the need for interpolating the displacement field. In [34], a motion tracking-based approach is presented, which can reconstruct the geometry of uniformly moving samples, with the sample motion being limited to the direction of the projected stripes. Additionally, the method requires the sample to have distinguishable features for motion tracking, which limits its applicability for texture-less samples.

The contribution of this paper is a novel method for SLP-based surface reconstruction that enables precise measurements of moving samples without the need to decode temporally-coded sequences, project additional patterns, rely on textured sample surfaces, or restrict the applicability to sample motion within the sub-pixel range. Surface reconstruction is achieved through an iterative optimization-based approach that simultaneously estimates the surface texture in terms of gray-scale intensities of surface points by directly processing the camera images and projection patterns while considering the motion of the sample. The procedure is designed for seamless integration into active stereo-vision systems and compatible with commonly used projection sequences.

This paper is organized as follows: Section 2 qualitatively outlines the challenges introduced by sample motion during multi-shot SLP. Section 3 presents the novel surface reconstruction algorithm and introduces the objective function which is minimized in an iterative process. Section 4 provides information on the experimental setup and parameters used for implementation. Finally, Section 5 evaluates the performance of the proposed approach in both static and dynamic scenarios.

2. Problem description

Conventional multi-shot SLP techniques rely on triangulation-based methods to reconstruct the geometry of a sample surface [35]. Triangulation refers to the geometric principle where the surface is reconstructed by determining the intersection of light rays emitted by the projector and detected by the camera. To establish pixel correspondences between the camera and the projector, the projector's pixel-columns are encoded through a sequence of projected images.

Two common encoding techniques utilize GC patterns and PSS patterns [36]. In the GC approach, each pixel-column of the projector is assigned a unique binary sequence consisting of alternating bright and dark values. This allows for robust identification of projector pixel-columns under various lighting conditions. On the other hand, PSS patterns encode pixel positions by the phase of the projected sinusoid. However, since the phase of a sinusoidal pattern is periodic, this leads to ambiguity in identifying the exact pixel-column position. Phase unwrapping can resolve these ambiguities for sufficiently flat, continuous surfaces. For discontinuous surfaces, phase-shifting techniques are frequently combined with auxiliary GC sequences, allowing for precise and unambiguous encoding [37].

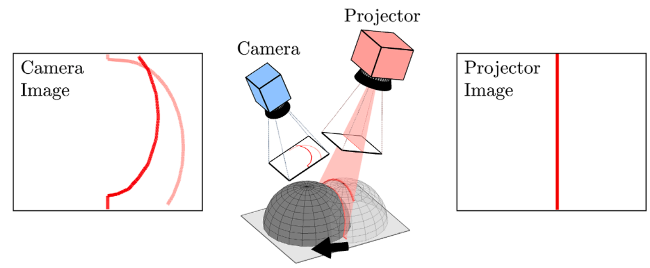


Fig. 1. A selected projector pixel-column is observed by the camera from a different point of view. Sample displacement, indicated by the black arrow, causes a displacement of the projector pixel-column in the camera image.

In static scenarios, the decoding process is straightforward. As sample and sensor do not move during the acquisition, a selected projector pixel-column is always found in the same location in the camera images. The camera captures the sequence of projected images, and each camera pixel is associated with a corresponding projector pixel-column by pixelwise decoding of the patterns [35]. This process enables accurate surface reconstruction, as each pixel correspondence is established without interference.

In scenarios where the sample moves between projections, the decoding faces a significant challenge. As shown in Fig. 1, the position of a selected projector pixel-column in the camera image is displaced due to the changing scene caused by the sample motion. The appearance of this displacement results from a combination of both the motion and the shape of the sample. For example, a flat sample (i.e., a plane) that moves in an in-plane manner will not cause any displacement of the projector pixel-columns in the camera images. The same motion with a semi-sphere mounted to the plane, such as depicted in Fig. 1, will cause a selected pixel-column to displace in the camera image. As the shape of the sample is unknown at the time of the measurement, the displacement of the pixel-column in the camera image is inherently unpredictable, even if the sample's motion is well-characterized. Consequently, a selected camera pixel observes different projector pixel-columns in different images, disrupting the decoding process and leading to erroneous correspondence assignments.

3. System and algorithm description

To bypass the challenge of correspondence assignment in SLP, the developed method introduces a novel approach for surface reconstruction that effectively handles samples that are moving in a known uniform manner without relying on pattern decoding. For this, an objective function that accounts for the correlation of the pixel values from the reprojected points in the images is constructed. An optimization procedure for iterative minimization of the objective function is developed.

The method is tailored but not limited to setups with one projector and two gray-scale cameras, as shown in Fig. 2a, projecting a sequence of images, such as depicted in Fig. 2b. The projector can be treated as an inverse camera, which enables the application of the same model to all components.

3.1. Camera and projector model

A pinhole camera model is applied to all cameras and the projector. Each component is calibrated such that camera intrinsics, camera extrinsics and distortion parameters are known, what enables to map 3D points from the world coordinate system to the image planes [38]. Coordinates in the world coordinate system are denoted as $[x, y, z]^{\mathrm{T}}$ while the 2D pixel location on the image plane is denoted as $[u, v]^{\mathrm{T}}$.

Camera Intrinsics

The intrinsic parameters describe the internal characteristics of the camera, which affect how the camera forms an image. These parameters define the reprojection of a 3D point onto the image plane, taking focal length, skew and the camera's principal point into account. The intrinsic matrix $\mathbf{K}$ is given as

$$\mathbf{K} = \begin{bmatrix} f_u & s & c_u \\ 0 & f_v & c_v \\ 0 & 0 & 1 \end{bmatrix}, \tag{1}$$

where f_u and f_v are the focal lengths in the u- and v-directions in pixels, c_u and c_v are the coordinates of the principal point (the optical center) on the image plane in pixels and s defines the skew between the axes of the image plane.

Camera extrinsics

The extrinsic parameters define the position and orientation of the camera in the world coordinate system. This allows to convert a 3D point from the world coordinate system to the camera coordinate system [38]. The camera extrinsics are composed of a 3×3 rotation matrix $\mathbf{R}$ and a 3×1 translation vector $\mathbf{t}$. A point $\mathbf{x}_w = [x_w, y_w, z_w]^T$ in world coordinates is transformed to camera coordinates $\mathbf{x}_c = [x_c, y_c, z_c]^T$ via

$$\mathbf{x}_c = \mathbf{R}\mathbf{x}_w + \mathbf{t}. \tag{2}$$

Distortion parameters

Distortion parameters allow to model distortions, caused by the camera lenses, which are not considered by the camera intrinsics. They are commonly described via radial distortion and tangential distortion. Radial distortion describes a symmetric distortion effect, where normalized image points $[\tilde{u}, \tilde{v}]^T$ shift towards or away from the principle point in radial direction. Tangential distortion describes an asymmetrical distortion effect caused by misalignment between the optical system and the image sensor. The distorted normalized image coordinates $[\tilde{u}_d, \tilde{v}_d]^T$ are calculated via

$$\tilde{u}_d = \underbrace{\tilde{u} + \tilde{u} \sum_{i=1}^{4} k_i (\tilde{u}^2 + \tilde{v}^2)^i}_{\text{Radial Distortion}} + \underbrace{\left(p_1(3\tilde{u}^2 + \tilde{v}^2) + 2p_2\tilde{u}\tilde{v}\right)}_{\text{Tangential Distortion}} \tag{3}$$

$$\tilde{v}_d = \underbrace{\tilde{v} + \tilde{v} \sum_{i=1}^{4} k_i (\tilde{u}^2 + \tilde{v}^2)^i}_{\text{Radial Distortion}} + \underbrace{\left(2p_1\tilde{u}\tilde{v} + p_2(\tilde{u}^2 + 3\tilde{v}^2)\right)}_{\text{Tangential Distortion}} \tag{4}$$

where k_1, k_2, k_3, k_4 are the radial distortion coefficients and p_1, p_2 are the tangential distortion coefficients [39].

3.2. Reprojection of 3D points onto the 2D image plane

The reprojection of a 3D point $\mathbf{x}_w = [x_w, y_w, z_w]^T$ from the 3D world coordinate system to the 2D image plane follows these steps:

3.2.1. Transformation from world coordinates to camera coordinates

Using the camera extrinsics, the transformed point is obtained as described in Eq. (2).

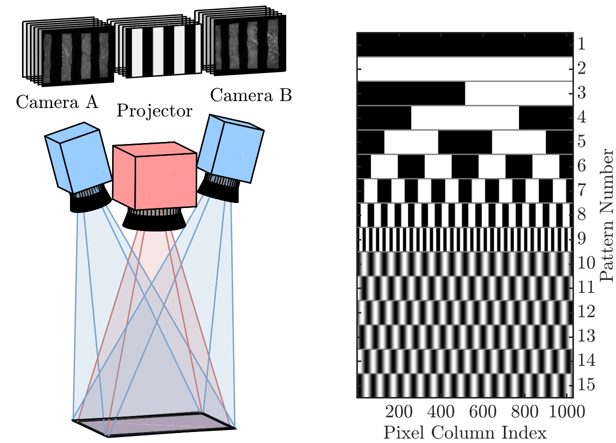
3.2.2. Perspective projection

The transformed points are reprojected to a virtual, normalized image plane via a central projection, resulting in normalized image points given by

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \end{bmatrix} = \begin{bmatrix} \frac{x_c}{z_c} \\ \frac{y_c}{z_c} \end{bmatrix}. \tag{5}$$

3.2.3. Correction of radial and tangential distortions

Distortion parameters are applied as described in Eqs. (3) and (4).



(a) Setup Schematic (b) Typical Projection Sequence

Fig. 2. The measurement system (a) for structured light profilometry (SLP) consists of a projector (red) that projects a sequence of patterns and two gray-scale cameras (blue) that simultaneously capture the projected patterns from a different perspective. A typical projection sequence (b) consists of a fully dark projection and a fully bright projection, followed by a Gray-coded (GC) binary sequence and multiple phase-shifted sinusoidal (PSS) patterns.

3.2.4. Reprojection onto the 2D image plane

Finally, the distorted points are reprojected to the 2D image plane using the intrinsic matrix of the camera yielding

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K} \begin{bmatrix} \tilde{u}_d \\ \tilde{v}_d \\ 1 \end{bmatrix}, \tag{6}$$

where u and v are the pixel coordinates of the reprojected 3D point in the image plane [40].

3.3. Correlation of pixel values

As described previously, any 3D point $[x, y, z]^T$ can be reprojected to the image triples (C^A, C^B, P) which are the captured images of camera A and camera B as well as the projected image at a certain time. In each image, the pixel values (p^A, p^B, p^P) of the reprojected point are retrieved using the obtained pixel coordinates from Eq. (6). Bi-linear interpolation is applied to estimate the pixel values between discrete pixel locations. If the considered 3D point is located at the surface of an opaque object, it is called a surface point of the sample. In this case, the retrieved pixel values (p^A, p^B, p^P) are expected to be similar: When the surface point lies within an illuminated region, the pixel values are relatively high, while in non-illuminated regions, they are lower. This similarity can be quantified by pairwise comparison of the pixel values using a correlation measure such as

$$c = \left(p^A - p^B\right)^2 + \left(p^A - p^P\right)^2 + \left(p^B - p^P\right)^2, \tag{7}$$

where values close to zero indicate high similarity of the pixel values. The pixel intensities of the camera images are not only defined by the brightness of the projected pattern, but also depend on the background illumination and the reflectivity of the sample surface (e.g., due to the surface finish, color and material). A commonly used model for phase-shifting SLP (PSP) applications [12] suggests that the intensity of a surface point I^C, as perceived by the point of view of the camera, is a result of the illumination I^P caused by the projection pattern and the illumination I^{BG} caused by diffuse background illumination and the dark level of the projector, described by

$$I^C = a\left(I^P + I^{BG}\right), \tag{8}$$

with a being the reflection coefficient modeling the surface reflectivity and color. From this, the projector illumination I^{P} can be calculated via

$$I^{\mathrm{P}} = \frac{1}{a} I^{\mathrm{C}} - I^{\mathrm{BG}}. \tag{9}$$

With the projector illumination I^{P} being proportional to the projector pixel value p^{P} and the camera perceived illumination I^{C} being proportional to the camera pixel values p^{A} and p^{B}, Eq. (9) allows the correction of the camera pixel values for the impact of the surface reflection and the background illumination via

$$\tilde{p}^{A} = \alpha^{\mathrm{A}} p^{\mathrm{A}} - \beta \quad \text{and} \quad \tilde{p}^{B} = \alpha^{\mathrm{B}} p^{\mathrm{B}} - \beta, \tag{10}$$

using the coefficients α^{A} and α^{B}, which relate to the reflectivity of the surface point, and β, which relates to the background illumination. For example, a dark surface leads to dark camera images. Choosing α^{A} and α^{B} accordingly high will correct for this impact. Intense background illumination leads to bright camera images, even if the projector is turned off. Increasing β will correct for that impact. Using separate reflection parameters for both cameras is necessary for the consideration of specular surface reflection, where light is reflected into the cameras with different intensities. Using the corrected camera pixel values in Eq. (7) yields the correlation measure

$$\begin{aligned} c =& \left(\alpha^{\mathrm{A}} p^{\mathrm{A}} - \alpha^{\mathrm{B}} p^{\mathrm{B}}\right)^2 \\ &+ \left(\alpha^{\mathrm{A}} p^{\mathrm{A}} - \beta - p^{\mathrm{P}}\right)^2 \\ &+ \left(\alpha^{\mathrm{B}} p^{\mathrm{B}} - \beta - p^{\mathrm{P}}\right)^2. \end{aligned} \tag{11}$$

3.4. Consideration of sample motion during acquisition

The set of sample surface points at an arbitrarily chosen position of the sample during its linear motion is represented by the point cloud $\mathcal{P}$, containing M points in Cartesian coordinates described by

$$\mathcal{P} = \left\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_M\right\}, \tag{12}$$

where

$$\mathbf{x}_m = \begin{bmatrix} x_m \\ y_m \\ z_m \end{bmatrix}, \quad m = 1, 2, \dots, M. \tag{13}$$

The measurement system simultaneously projects and captures N patterns in regular time intervals, obtaining N image triples $(C_n^{\mathrm{A}}, C_n^{\mathrm{B}}, P_n)$ with $n = 1, 2, \dots, N$. Due to uniform linear motion, the sample experiences a shift of $\mathbf{d} = [d_{\mathrm{x}}, d_{\mathrm{y}}, d_{\mathrm{z}}]^{\mathrm{T}}$ between two subsequent captures. Hence, all surface points of the rigid sample are shifted equally by the translation vector $\mathbf{d}$. The resulting shifted point cloud $\mathcal{P}_n$ that represents the set of sample surface points at the acquisition time of the image triple $(C_n^{\mathrm{A}}, C_n^{\mathrm{B}}, P_n)$ is

$$\mathcal{P}_n = \left\{\mathbf{x}_{1,n}, \mathbf{x}_{2,n}, \dots, \mathbf{x}_{M,n}\right\} \tag{14}$$

where each shifted point $\mathbf{x}_{m,n}$ can be calculated from the point cloud $\mathcal{P}$ via

$$\mathbf{x}_{m,n} = \mathbf{x}_m + \left(n - n_0\right)\mathbf{d}. \tag{15}$$

The parameter n_0 can be chosen arbitrarily and defines the position of the point cloud $\mathcal{P}$ along the trajectory of the sample during the acquisition. For example, if n_0 is chosen to be 1, $\mathcal{P}$ is congruent with $\mathcal{P}_1$. It makes sense to place $\mathcal{P}$ in the center of the sample trajectory by selecting n_0 to be $N/2$. In this case, the mean of all absolute displacements between the point clouds $\mathcal{P}_n$ and $\mathcal{P}$ is minimal.

For each point cloud $\mathcal{P}_n$ and each point $\mathbf{x}_{m,n}$ the pixel values $(p_{m,n}^{\mathrm{A}}, p_{m,n}^{\mathrm{B}}, p_{m,n}^{\mathrm{P}})$ are retrieved from the corresponding image triples $(C_n^{\mathrm{A}}, C_n^{\mathrm{B}}, P_n)$. Finally, the correlation measure from Eq. (11) is evaluated for each triple of pixel values yielding the objective function

$$\begin{aligned} c(z_m, \alpha_m^{\mathrm{A}}, \alpha_m^{\mathrm{B}}, \beta) =& \sum_{n=1}^{N}\sum_{m=1}^{M} \left(\alpha_m^{\mathrm{A}} p_{m,n}^{\mathrm{A}} - \alpha_m^{\mathrm{B}} p_{m,n}^{\mathrm{B}}\right)^2 \\ &+ \sum_{n=1}^{N}\sum_{m=1}^{M} \left(\alpha_m^{\mathrm{A}} p_{m,n}^{\mathrm{A}} - \beta - p_{m,n}^{\mathrm{P}}\right)^2 \\ &+ \sum_{n=1}^{N}\sum_{m=1}^{M} \left(\alpha_m^{\mathrm{B}} p_{m,n}^{\mathrm{B}} - \beta - p_{m,n}^{\mathrm{P}}\right)^2, \end{aligned} \tag{16}$$

where each surface point is assigned its own reflectivity parameters α_m^{A} and α_m^{B}. The choice of constant reflectivity parameters over all N captured images for each camera implies Lambertian reflection characteristics within the limits of slight variations in the relationship between the surface normal, the projection angle and the viewing angle of a particular surface point during sample movement. As the background illumination is assumed to illuminate each surface point equally, a global background illumination parameter β is used. Alternatively, β can be assigned individually for each point. In this formulation, every 3D point is associated with four optimization variables, and the objective function in Eq. (16) decomposes into multiple independent optimization problems with a total of $4M$ variables. Introducing a single global parameter β reduces the number of optimization variables to $3M + 1$, but at the expense of coupling the points. This coupling increases memory demands during computation, particularly for the Hessian matrices.

Optionally, to promote correct phase assignment when using the optimization-based method without auxiliary patterns for unambiguous phase assignment, an additional term

$$c_{\mathrm{a}}(z_m) = w_{\mathrm{a}} \sum_{(i,j)\in\mathcal{I}} \left(z_i - z_j - \tilde{z}_{i,j}\right)^2, \tag{17}$$

which assesses the difference of the z-coordinates of adjacent points to penalize high gradients in the reconstructed surface, can be considered in the cost function. In this term, w_{a} represents the weight, $\mathcal{I} = \{(i_1, j_1), (i_2, j_2), \dots\}$ is the set of index pairs, which define adjacent points and $\tilde{z}_{i,j}$ is the expected height difference between the points $\mathbf{x}_i$ and $\mathbf{x}_j$. This height difference can be chosen to be zero for smooth samples or calculated from initial values for z_m.

The objective function is used in an optimization framework in order to reconstruct the geometry of the moving sample by finding the global minimum.

3.5. Optimization-based surface reconstruction

The proposed method utilizes the objective function defined in Eq. (16) for surface reconstruction. The evaluation of this function requires the following information:

- calibration parameters for camera A, camera B and the projector
- image triples with pixel values in the range from 0 to 1
- vector of sample motion between images $\mathbf{d}$
- surface points $\mathcal{P}$
- reflectivity parameters α_m^{A} and α_m^{B}
- background illumination parameter β

Calibration parameters, including extrinsics, intrinsics and distortion parameters, are determined using camera calibration [41]. Image triples are available after the data acquisition. The uniform sample motion is assumed to be known. The sample surface points are evaluated on a fixed grid defined by the x- and y-coordinates. Hence, the depth information (i.e., the z-coordinates z_m of the surface points), the reflectivity parameters α_m^{A} and α_m^{B} and the background illumination β are unknown. This set of unknown parameters $\left\{z_m, \alpha_m^{\mathrm{A}}, \alpha_m^{\mathrm{B}}, \beta | m = 1, 2, \dots, M\right\}$ is estimated by finding the global minimum of the objective function in Eq. (16).

Considering a single 3D point with predefined x- and y-coordinates, and suitably chosen reflectivity and background illumination parameters, the cost term that contributes to the objective function can be plotted in dependency of the z-coordinate, as shown by the green line in

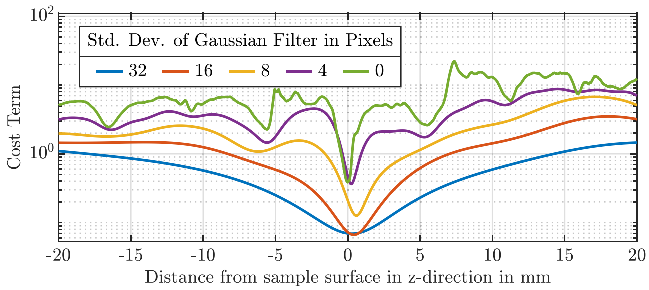


Fig. 3. The objective function in dependency of the distances of a single 3D point to the sample surface in z-direction appears smoother for low-pass filtered images. A Gaussian filter with the given standard deviation is used for filtering.

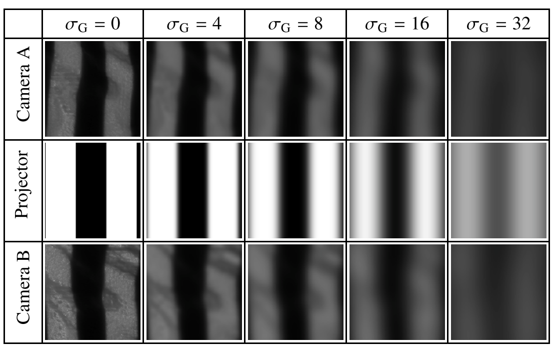


Fig. 4. Image sections of the projected and captured images for different levels of filtering, defined by the standard deviation σ_G of the Gaussian low-pass filter.

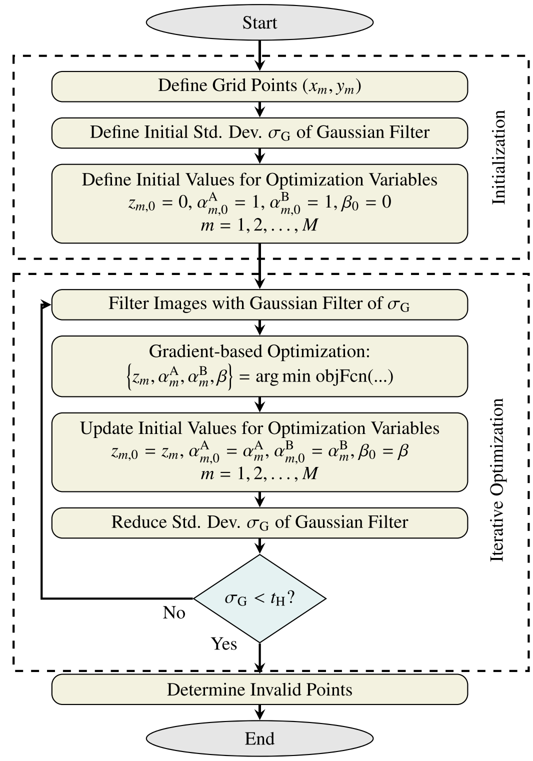


Fig. 5. Flowchart of the reconstruction procedure. Throughout the iterations of the optimization process, the filtering level decreases and the approximated objective function as well as the determined global minimum approach the original objective function and its global minimum.

Fig. 3. As the z-coordinate is varied, the 3D point moves along a line perpendicular to the $x-y$-plane. The reprojections of the 3D point into the image triples also move along a line, causing the triples of pixel values to change constantly. The high intensity gradients in the projected images lead to steep gradients in the cost term, resulting in a rapidly varying objective function with a large number of local minima. This diminishes the chances of a successful gradient-based minimum search. The application of low-pass filters to the image triples creates a smooth approximation of the objective function and its global minimum. Filtering is performed using a Gaussian image filter, where the standard deviation σ_G of the Gaussian filter kernel defines the bandwidth and thus the level of smoothing. Fig. 4 shows a section of an image triple, processed with Gaussian low-pass filters with different standard deviations. To ensure comparable levels of filtering between the captured and projected images, the standard deviation of the kernel applied to the captured camera images is adjusted based on the resolution ratio between the captured and projected images.

Fig. 3 illustrates the cost term for a single 3D point as a function of the vertical distance to the actual surface for different levels of filtering. As the standard deviation of the Gaussian filter decreases, the approximated objective functions become progressively less smooth, converging towards the unfiltered objective function. Despite the decreasing smoothness, the global minimum is consistently approximated as a distinct valley. This observation supports the use of an iterative optimization strategy, starting from a high level of filtering and gradually reducing the standard deviation of the filter in subsequent iterations. The minimum determined in each iteration is used as an initial value for the following iteration, ensuring a reliable approximation to the global minimum of the unfiltered objective function.

The procedure is summarized in Fig. 5. The iterative reconstruction terminates when the standard deviation σ_G falls below a threshold t_H. After that, invalid points are determined based on the following criteria:

3.5.1. *Reprojection beyond image borders*

Points that are reprojected beyond the image borders of any image of the image triples are considered invalid, as the missing information reduces the accuracy of the result.

3.5.2. *High cost values*

Points that contribute an exceptionally high value to the objective function indicate a lack of similarity between the triples of pixel values. This may be due to a wrong estimation of the z-coordinates or due to occlusion and shadowing effects caused by the different points of view for the cameras and the projector.

3.5.3. *High distance to surrounding surface points*

Points that exhibit high distances to surrounding surface points signal noisy points and outliers, which are considered invalid.

3.5.4. *Unreasonable estimated gray-scale intensity*

The gray-scale intensities I^A_m and I^B_m of a surface point perceived by the cameras for a fully illuminated projection image can be estimated from the reflectivity and background illumination parameters. For fully illuminated surface points, the pixel values in the projected image are 1 and the corresponding corrected pixel values in a captured image would be $\alpha^A_m I^A_m - \beta$ and $\alpha^B_m I^B_m - \beta$ (c.f. Eq. (11)). Hence, the camera-perceived gray-scale intensities are calculated by

$$I^A_m = \frac{1+\beta}{\alpha^A_m} \quad \text{and} \quad I^B_m = \frac{1+\beta}{\alpha^B_m}. \tag{18}$$

Points with an associated gray-scale intensity greater than 1 indicate that the camera-perceived intensity exceeds the projected intensity. This can occur in surface areas with strong specular reflection components when gray-scale images, such as sinusoidal patterns, are projected or due to incorrect estimation of the z-coordinates. These points are thus also considered invalid.

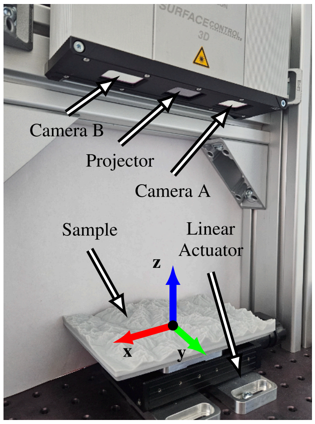


Fig. 6. The experimental setup consists of two cameras and a projector which are housed in an industrial sensor (Micro Epsilon surfaceControl3500-120), as well as a linear actuator which moves the sample. The sample is a 3D printed model of a section of the Austrian Alps, including the Großglockner, the highest mountain in Austria.

The proposed method regards the reconstruction method subsequent to the data acquisition. Hence, it can be integrated into any SLP setup without requiring modifications to the hardware or acquisition procedures. It retains compatibility with conventional multi-shot projection sequences, such as GC and PSS patterns, making it a versatile approach which potentially upgrades the functionality of SLP systems towards surface reconstruction of both static and moving samples.

4. Implementation

The experimental setup consists of an industrial SLP sensor (Micro Epsilon surfaceControl3500-120), a linear actuator (PI VT80-50DC) and a 3D printed sample, as shown in Fig. 6. The sensor houses a projector and two gray-scale cameras. As already depicted in Fig. 2a, the field of view of the cameras matches the projection size of the projector. The cameras capture gray-scale images with pixel values in the range 0 to 1. The projector projects gray-scale images with pixel values in the range 0 to 1. The sequence of pattern is similar to the sequence depicted in Fig. 2b and consists of a dark image (no illumination) followed by a bright image (fully illuminated), a set of 8 GC images and 16 PSS images with a period of 24 pixels. The linear actuator displaces the sample between consecutive patterns.

The presented procedure is implemented in MATLAB using the trust-region optimization algorithm from the Optimization Toolbox [42]. Gradients and Hessian of the objective function are calculated analytically for faster convergence. In order to smooth the objective function sufficiently, the standard deviation σ_G for the Gaussian low-pass filter applied to the projector images of the first iteration is chosen to be 36 pixels, which exceeds the period of the high-frequency patterns in the projection sequence and therefore efficiently suppresses these frequencies throughout the first iterations. The standard deviations of the ath iteration is $40 \cdot 0.9^a$. For the Gaussian filter of the camera images, this standard deviation is multiplied by a factor of 1.13, what results in similar filtering level considering the different image resolutions. In order to counteract the effects of minor calibration errors and camera noise, the optimization is stopped when σ_G drops below $t_H = 1$ (after 35 iterations). The size of the square Gaussian filter kernel is defined to be $2 \cdot \lceil 3\sigma_G \rceil + 5$, with $\lceil \cdot \rceil$ being the ceiling function.

As indicated in Fig. 5, the initial values for the optimization are chosen to be zero for all z-coordinates, which corresponds to the center of the measurement range of the sensor, where both cameras and the projector have their focal planes. The α^A and α^B parameters are initialized with one while β is initialized with zero, which corresponds to a situation without background illumination and uniform diffuse reflectivity of the sample surface.

Table 1
Sample motion between consecutive projected patterns for different testcases.

	x	y	z
Testcase 1	0.0 µm	0.0 µm	0.2 µm
Testcase 2	−1.4 µm	299.9 µm	1.3 µm
Testcase 3	−300.0 µm	−2.6 µm	1.3 µm
Testcase 4	−130.8 µm	150.5 µm	20.9 µm

5. Experimental results

The performance of the optimization-based reconstruction approach is evaluated for various directions of uniform motion and the results are compared to those obtained by a conventional decoding-based reconstruction approach [43]. The conventional reconstruction approach processes the same images as the optimization-based method and also enables the reconstruction on a predefined grid while taking advantage of both cameras. The z-coordinates are obtained by reprojecting the 3D points into the unwrapped phase maps of both cameras, which are obtained by pattern decoding [44]. Four testcases are defined as summarized in Table 1, where the sample is displaced in different directions between consecutive images during the acquisition phase. Testcase 1 serves as reference, i.e., a static scenario without motion. Testcase 2 investigates motion in y-direction, corresponding to a movement orthogonal to the projected stripes. In Testcase 3, dominant motion occurs in x-direction, i.e., parallel to the projected stripes. Testcase 4 considers a general case, with translational motion components in all directions. The motion direction relative to the world coordinate system is determined beforehand using two static measurements acquired with the sample positioned 20 mm apart along the actuator's trajectory. A rigid-body transformation between the two resulting point clouds is then estimated using Iterative Closest Point (ICP) registration, which directly provides the translation direction [45].

During data acquisition, the SLP sensor projects and captures images every second while the linear actuator moves the sample step by step while using the internal encoder to monitor the sample position. This stop-and-go acquisition minimizes the occurrence of synchronization uncertainties or position errors. The acquired image triples are processed with both the novel optimization-based reconstruction method and the conventional decoding-based reconstruction method. The ground truth is obtained using the original software of the industrial sensor to measure the static sample. Both results are compared to this ground truth using ICP registration. The RMS of the residual Euclidean distances between the points of the aligned point clouds and the reference surface is used as a numerical measure to evaluate the accuracy of the measurement [46].

The surface points are reconstructed on an equidistant grid of x- and y-coordinates with a grid period of 500 µm covering a range from −40 mm to 40 mm for the x-coordinates and −30 mm to 30 mm for the y-coordinates, with the z-coordinate and the texture to be estimated during the optimization.

For the optimization-based approach, invalid points are determined as described in Section 3.5. The thresholds are selected empirically based on typical values of the objective function and the geometric considerations. A point is considered invalid if its average correlation measure per image triple (c.f., Eq. (11)) exceeds 0.06 and therefore the contribution to the overall cost exceeds a threshold of approximately 1.5, which reflects the typical variability observed for correctly reconstructed points. The neighborhood-distance threshold is guided by the spatial sampling of the reconstructed surface (500 µm) and by the triangulation geometry of the sensor. At this sampling density, a point located more than 2 mm from its neighbors would necessarily be occluded

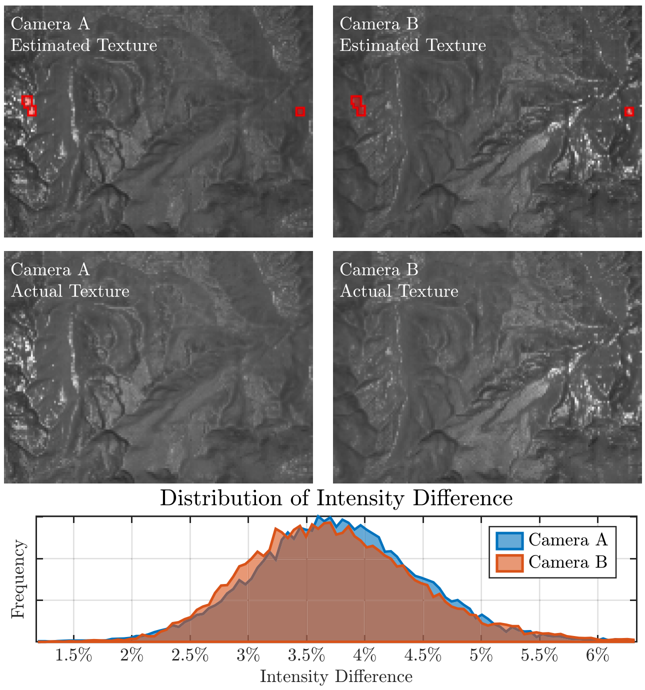


Fig. 7. The estimated texture results from the estimated parameters α_m^{A}, α_m^{B} and β. The comparison to the actual surface texture retrieved from the fully illuminated projection shows a mean difference of about 3.7 %. Invalid points are marked with red boxes.

Table 2
Relative intensity difference between the reference texture image and the estimated texture image.

	Intensity Difference in Camera A		Intensity Difference in Camera B	
	Mean	Std. Dev.	Mean	Std. Dev.
Testcase 1	3.8 %	0.9 %	3.7 %	0.8 %
Testcase 2	3.7 %	1.2 %	3.8 %	1.4 %
Testcase 3	3.7 %	1.2 %	3.9 %	1.5 %
Testcase 4	4.2 %	0.8 %	4.2 %	1.5 %

or occluding, making such points physically implausible. Hence, points with a median of the Euclidean distance to the eight nearest neighbors of more than 2 mm are considered invalid. Contrary to Section 3.5.4, estimated gray-scale intensities of up to 1.05 are tolerated, allowing for minor numerical deviations.

For the conventional approach, a point is considered invalid if the phase values of the reprojected point in both unwrapped phase images of the cameras differ by more than $\sqrt{10}$ projector pixels or if the median of the Euclidean distance to the eight nearest neighbors is below 2 mm.

5.1. Estimation of texture and gray-scale intensity

The optimization estimates the reflectivity parameters α_m^{A} and α_m^{B}, as well as the background illumination parameter β. Using these parameters, the camera-perceived gray-scale intensities of the sample surface points, and thus the texture of the surface, can be reconstructed via Eq. (18). Fig. 7 shows the estimated texture of the surface for both cameras and compares the result to the actual texture. The actual texture is retrieved from a camera image of the sample being uniformly illuminated by the projector. The estimated surface points are reprojected into this camera image in order to get the actual texture for those points. The histogram depicts the distribution of intensity difference relative to the range of the images, which is 1. The distributions for both cameras are similar, exhibiting a mean of approximately 3.7 % and a standard deviation of approximately 0.8 %. The positive mean values indicate that the actual texture image is brighter than the estimated texture image, which can be explained by inter-reflections and the contrast characteristics of the projector. A single pixel appears brighter when surrounding pixels are turned on, which causes decreased brightness for stripe patterns as compared to the fully illuminated projection. Additionally, the models used to derive the objective function assume spatially and temporally uniform background illumination. This assumption is supported by the measurement conditions: the setup was covered during acquisition to block external light. Therefore, stray light caused by diffuse reflections of the projected light from the sample surface, followed by secondary reflections from the surrounding hardware, is the main contributor to the background illumination. The mean image intensity across each projected pattern is similar for all fringe patterns (approximately 0.5), indicating a stable ambient offset. For the fully illuminated pattern, the mean image intensity is higher, leading to increased background illumination, which also justifies the higher mean intensity of the actual texture image. This, together with the inhomogeneous distribution of the projected illumination in the fringe patterns and the spatially varying reflectance of both the sample and the surrounding hardware, introduces slight deviations in the background illumination, which are not accounted for by the objective function.

Table 2 summarizes the mean and the standard deviation of the intensity difference for each testcase. The standard deviation of less than 2 % suggests a good estimation of the actual texture. Testcases with sample motion exhibit higher standard deviations as compared to the static scenario in testcase 1. This results from the choice of constant parameters α_m^{A} and α_m^{B} during the acquisition. However, when the sample is moving, the camera-perceived gray-scale intensity can change due to specular reflections. The effect on the reconstruction result might increase for samples with highly reflective surfaces. As these surfaces pose a general performance limitation of SLP, they are not further of interest for this study.

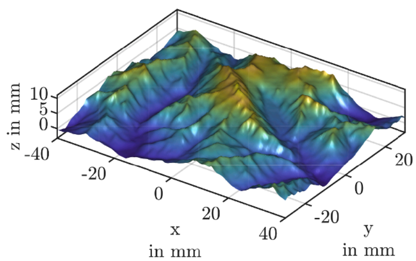
5.2. Surface reconstruction

Fig. 8 shows the results of the surface reconstruction for static and moving samples. Missing points are those classified as invalid. The application of the conventional reconstruction method to static scenarios (Fig. 8a) obtains a decent result. The same reconstruction method applied to moving samples (Fig. 8b) leads to an inaccurate result with a high number of invalidly reconstructed points. In contrast, the optimization-based reconstruction approach applied to the moving sample (Fig. 8c) is able to obtain a similar result as in the static scenario.

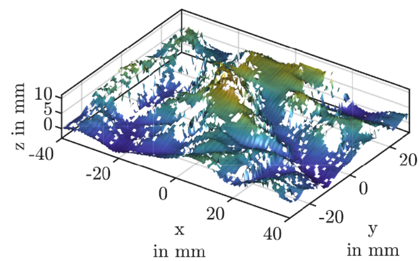
In Table 3, the results are expressed quantitatively, comparing the percentage of invalid points as well as the RMS of the residual distance between the ICP registered point clouds and the ground truth surface. The results show that the conventional measurement performs well for the static scenario, yielding an RMS error of 11.0 µm. The optimization-based approach performs similarly, with an RMS error of 17.2 µm. For reference, the RMS error between two ground truth measurements acquired by the original software of the industrial sensor is 12.2 µm. The superiority of the optimization-based algorithm is evident for moving samples, where the residual errors in the conventional approach increase significantly by a factor of more than 20, while remaining unchanged for the optimization-based method. Additionally, the number of invalid points in the conventional approach increases with larger shifts, whereas it remains almost negligible for the optimization-based approach.

5.3. Impact of motion errors

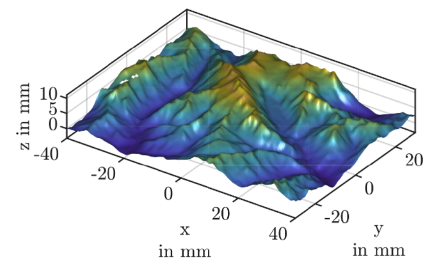
The surface reconstruction demands well-characterized linear sample motion. In practice, geometrical motion errors, calibration errors, synchronization uncertainties or vibration cannot be ruled out. To estimate the impact of these errors on the reconstruction results, a positioning error $\mathbf{e}_n = [e_{\mathrm{n,x}}, e_{\mathrm{n,y}}, e_{\mathrm{n,z}}]^{\mathrm{T}}$ for each sample position at the acquisition



(a) Static Measurement



(b) Measurement in Motion (Conventional Approach)



(c) Measurement in Motion (Optimization-based Approach)

Fig. 8. Results of the surface reconstruction for (a) a static measurement, (b) a measurement in motion using the conventional decoding-based reconstruction approach and (c) a measurement in motion using the novel optimization-based reconstruction approach. Missing surface points are considered invalid due to high gradients or discrepancies between the images of both cameras.

Table 3
Results of the surface reconstruction for different approaches.

	Conventional Approach Percentage of Invalid Points	RMS Residual Error	Optimization-based Approach Percentage of Invalid Points	RMS Residual Error
Testcase 1	0.0 %	11.0 µm	0.03 %	17.2 µm
Testcase 2	12.9 %	492.9 µm	0.06 %	17.7 µm
Testcase 3	15.3 %	539.4 µm	0.03 %	16.6 µm
Testcase 4	7.8 %	268.8 µm	0.03 %	17.4 µm

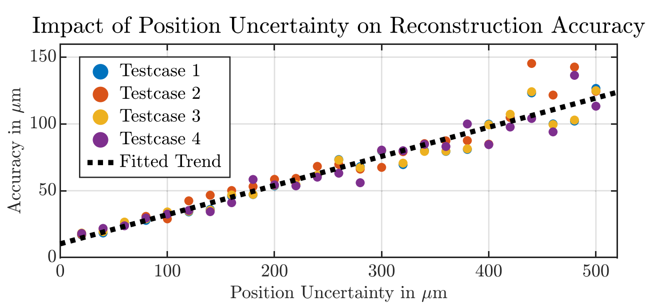


Fig. 9. Position uncertainties due to calibration errors or geometrical motion errors decrease the accuracy of the reconstruction results.

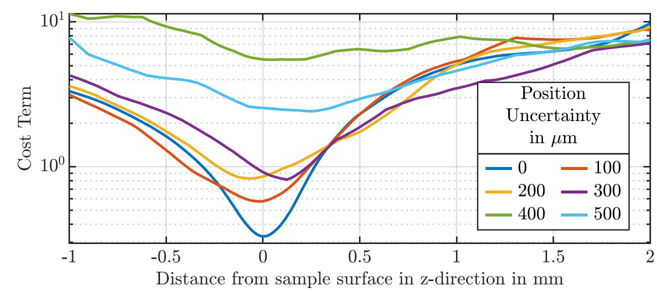


Fig. 10. In the presence of position uncertainties, the global minimum of the objective function, evaluated along the distance of a selected 3D point to the sample surface in z-direction, shifts from its original position and is less pronounced.

Table 4
Statistics of the reconstruction result from 50 optimization runs with different initial values.

	Testcase 1	Testcase 2	Testcase 3	Testcase 4
Standard Deviation of Reconstructed Surface	0.08 µm	0.07 µm	0.07 µm	0.13 µm
Relative Deviation of Final Cost Value	0.001 %	0.000 %	0.000 %	0.002 %
Max. Percentage of Invalid Points	0.06 %	0.06 %	0.04 %	0.23 %

time of the nth image triple is simulated by distorting the linear trajectory of the sample motion, which is considered in the objective function as defined in Eq. (15), via

$$\mathbf{x}_{m,n} = \mathbf{x}_m + (n - n_0)\mathbf{d} - \mathbf{e}_n. \tag{19}$$

For all testcases, various positioning errors are evaluated by assuming a normally distributed position uncertainty with different standard deviations between 20 µm and 500 µm. Fig. 9 depicts the accuracy of the reconstruction results. The decrease in accuracy of only 22 µm per 100 µm of additional position uncertainty indicates that the proposed method is capable of providing good reconstruction results in the presence of position uncertainties. Same as the green line in Figs. 3, 10 shows the contribution of a selected point to the objective function, in dependence of the z-coordinate for different position uncertainties. The higher discrepancy between the retrieved pixel values in the presence of positioning errors causes the location of the global minimum to become increasingly distant from the original location and less pronounced. Importantly, the global minima are still well-distinguishable from the local minima, enabling robust convergence.

5.4. Impact of initial values

To inspect the reliability of the global minimum and the robustness of the conversion, the iterative optimization process is started several times with different initial values. All points get assigned the same random value between −15 mm and 15 mm for their z-coordinates, which relates to the specified measurement range of the industrial sensor, and common random value between 1 and 4 for all parameters α_m^A and α_m^B, which is considered the typical range of these parameters. The parameter β is initialized with a random value between 0 and 0.2. Table 4 summarizes the statistics for 30 optimization runs with different initial values for each testcase. The standard deviation of the reconstructed surface measures the mean standard deviation of all estimated z-coordinates and can be interpreted as the repeatability of the reconstruction process when using various initial values but the same image set. The value of less than 0.15 µm, which is about 100 times less than the determined accuracy, clearly indicates the robustness of the iterative optimization. The optimization process converges towards the same minimum, with final costs that deviate by less than 0.002 %, as compared to its absolute value. Also, the percentages of invalid points are stable and comparable to the initialization strategy described in Section 4.

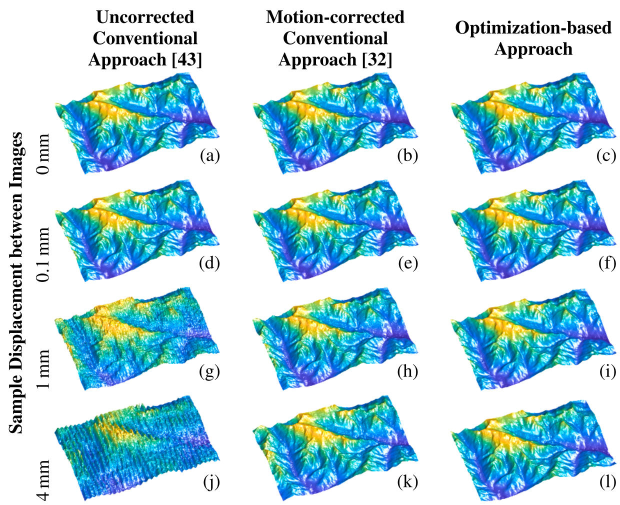


Fig. 11. Reconstruction results from different methods using three-step phase-shifting SLP (3-PSP).

Table 5
Reconstruction accuracy using 3-PSP.

Sample Displacement between Images	Uncorrected Conventional Approach [43]	Motion-corrected Conventional Approach [32]	Optimization-based Approach
0 µm	30.8 µm	30.8 µm	37.5 µm
10 µm	34.3 µm	31.1 µm	26.4 µm
20 µm	41.8 µm	32.0 µm	27.2 µm
50 µm	66.6 µm	32.5 µm	27.7 µm
100 µm	100.8 µm	32.6 µm	27.9 µm
300 µm	183.7 µm	35.4 µm	30.8 µm
500 µm	221.7 µm	40.2 µm	34.9 µm
1000 µm	297.0 µm	65.4 µm	50.3 µm
2000 µm	438.9 µm	1277.7 µm	1811.8 µm
4000 µm	729.6 µm	71.4 µm	61.2 µm

5.5. Comparison to state-of-the-art

In this section, the proposed method is evaluated using three-step phase-shifting SLP (3-PSP), which reconstructs the sample surface from three projected PSS patterns. This allows for a direct comparison with state-of-the-art motion-corrected methods that utilize a reduced number of projected PSS patterns, without relying on auxiliary binary patterns for unambiguous phase assignment. In order to comply with the resulting limitations and enable phase unwrapping [19], the pattern period is increased and chosen to be 96 pixels for this experiment. Due to the missing auxiliary patterns and the associated ambiguity of the repeating phase, the term according to Eq. (17) is included in the objective function with a weight of $w_a = 10\sigma_G^2$, which supports the convergence towards to correct minimum. The acquired image sets, each containing three projection patterns, are processed for surface reconstruction using three different approaches:

1. Conventional Approach: The original captured images are used for the calculation of the wrapped phase map. The unwrapped phase map is calculated using a spatial phase unwrapping algorithm [47] and the surface is reconstructed using a conventional decoding-based method [43].
2. Motion-corrected Approach: The phase map is corrected using a motion-correction algorithm [32]. The surface is reconstructed using the corrected phase map and a conventional decoding-based method [43].
3. Optimization-based Approach: The surface is reconstructed using the proposed method.

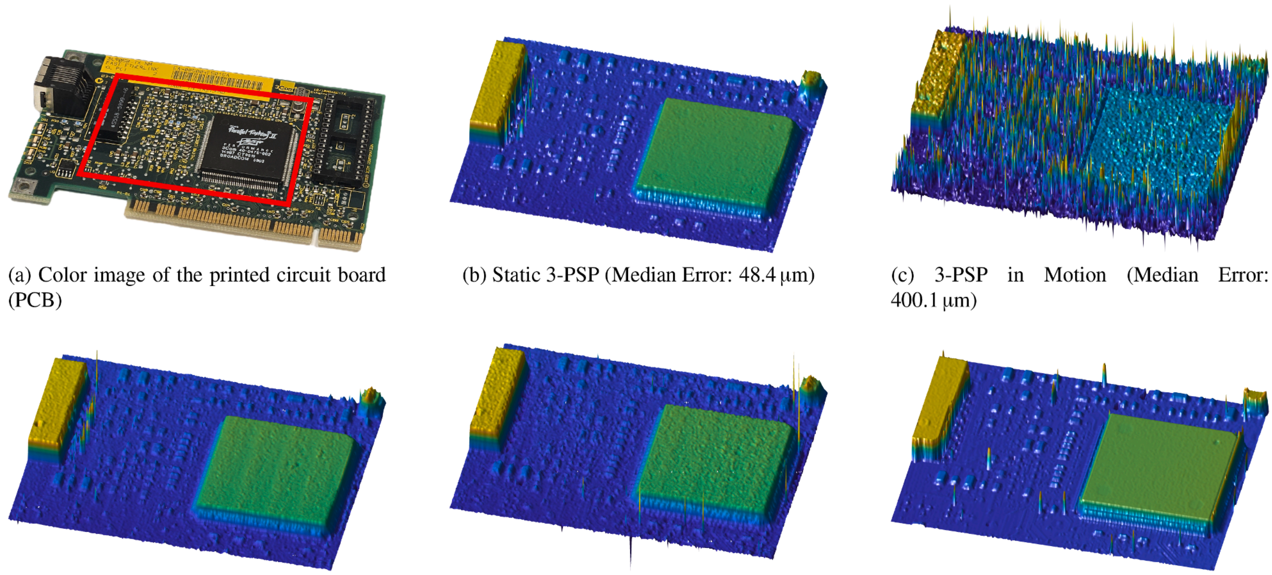
For this experiment, the sample is displaced linearly in the x-direction, which introduces maximal phase errors [34]. The sample motion between consecutive projected patterns is varied between 0 mm and 4 mm. All methods process the same image set and utilize information from both cameras. Both the motion-corrected approach [32] and the proposed method process the sample motion, which is assumed to be known. Additionally, the motion-corrected approach requires a coarse initial reconstruction result, which is provided by the conventional uncorrected approach.

Fig. 11 compares the results from the different reconstruction methods qualitatively. The conventional uncorrected reconstruction shows severe motion-induced ripples for large sample displacements. Both the motion-corrected conventional reconstruction and the optimization-based reconstruction yield good results, which are comparable to those of the static conventional 3-PSP. Table 5 quantitatively compares the results for various sample displacements between consecutive projected patterns. Again, the reconstruction accuracy of the optimization-based method remains consistent across all experiments, whereas the accuracy of the conventional method decreases as the sample displacement increases. The reduced accuracy of the optimization-based approach in comparison to previous experiments is attributed to the reduced number of projected patterns and the increased fringe period. On the other hand, the increased fringe period improves tolerance of the conventional approach to sample motion, resulting in higher accuracy for small sample displacements compared to the previous experiment. The motion-corrected conventional reconstruction method yields good results, albeit slightly less accurate than the proposed optimization-based method, which can be attributed to the linearized pixel error models and the coarse reference required by this method. The decreasing accuracy with increasing sample displacement is attributed to the progressive loss of phase information due to the reduced perceived phase shift as the sample moves together with the projected fringes. This leads to a decreased SNR in locations that consistently receive low illumination and culminates in a total loss of phase information for the experiment, with sample displacement of 2 mm, where the physical motion matches the phase shift of the PSS pattern in the focus plane.

5.6. Complex samples in motion

Finally, a complex, industrially relevant sample, a printed circuit board (PCB), is measured in a dynamic experiment, where the PCB is displaced by 300 µm in the x-direction between consecutive projections. As depicted in Fig. 12a, the PCB exhibits a rich texture, multiple highly specular reflective areas, and enhanced features, which lead to shading and occlusion effects. Both previously used projection sequences, the combined GC + PSS sequence and the 3-PSP sequence, are evaluated. Fig. 12b to f illustrate the reconstruction results obtained without removing potentially invalid points. The conventional 3-PSP method produces smooth and accurate reconstructions in the static case (Fig. 12b). However, when the sample is moved during acquisition, the result exhibits several obvious artifacts and erroneous points (Fig. 12c). In comparison to the previously used model of the Austrian Alps, the motion-induced errors are significantly more pronounced due to the stronger textural variations in the PCB images, which increase pixel-wise intensity fluctuations and amplify phase-decoding errors.

Both the corrected and the optimization-based 3-PSP reconstruction substantially mitigate these errors (Fig. 12d and e). However, remaining artifacts are visible, particularly around highly reflective components and dark regions (e.g., at the large chip, represented by the green surface). To quantify the reconstruction quality, the median point-to-plane distance is calculated with respect to the registered ground truth. Using the median value allows for the suppression of the impact of occasional outliers and less meaningful distance values for points at steep edges, where the point cloud density is low, or the reconstruction is affected by shading or occlusion. Comparing the accuracies of 58.5 µm for the corrected method and 86.8 µm for the optimization-based method to the



(a) Color image of the printed circuit board (PCB)

(b) Static 3-PSP (Median Error: 48.4 μm)

(c) 3-PSP in Motion (Median Error: 400.1 μm)

(d) Corrected 3-PSP in Motion (Median Error: 58.5 μm)

(e) Optimization-based 3-PSP in Motion (Median Error: 86.8 μm)

(f) Optimization-based SLP in Motion using GC+PSS patterns (Median Error: 21.2 μm)

Fig. 12. Surface reconstruction of a complex sample featuring rich texture, specular reflective areas, and enhanced geometric features.

result of the uncorrected reconstruction with an accuracy of 400.1 μm shows the effectiveness of both approaches. However, the highly specular reflective surface areas and the sparse amount of information resulting from the reduced number of patterns hinder the optimization-based reconstruction process, limiting its performance.

Importantly, the optimization-based approach offers high versatility and can incorporate projection sequences of arbitrary length. When applied to the combined sequence containing 8 GC patterns and 16 PSS patterns, the optimization-based reconstruction achieves a substantially improved accuracy of 21.2 μm, while producing visually smooth results, most notably across the surface of the large chip, represented by the green area in Fig. 12f. Occasional remaining outliers are primarily associated with regions of strongly specular reflecting surfaces.

In summary, the experiments demonstrate that the proposed algorithm is capable of robustly reconstructing the surface and texture of various static and moving samples with consistent accuracy while being applicable to different projection sequences.

6. Conclusion

The presented SLP method achieves surface reconstruction by finding the global minimum of an objective function that accounts for the similarity of pixel values in a set of images. For this, an iterative optimization procedure was developed that approximates the objective function utilizing image low-pass filtering methods, which ensures robust convergence to the global minimum using gradient-based solvers. This optimization-based approach can reconstruct the surface of uniformly moving samples in any direction with an accuracy of about 17 μm, which outperforms the results from conventional reconstruction by a factor of more than 20 for dynamic testcases while obtaining comparable results for static scenarios. The gray-scale intensities of the surface points can be reconstructed from the reflectivity parameters α_m^{A} and α_m^{B} and the background illumination parameter β with deviations to an actual capture of the surface texture of less than 2 %. The performance has been shown for a commonly used sequence of projected patterns using an industrial SLP sensor, which emphasizes the possibility of seamless integration into existing SLP systems, however, the algorithm can be applied to various projection sequences, as demonstrated using a 3-PSP sequence. The experiments demonstrate good robustness against position uncertainties of the sample. However, the use of a simplified reflection model employing the parameters α_m^{A}, α_m^{B}, and β leads to limitations when measuring moving samples with highly reflective surfaces or complex background illumination, which would demand a more sophisticated model. In addition, real-time capability requires powerful computing units or further efforts for efficient implementation due to the computationally intensive iterative optimization.

The performance of the presented method was demonstrated for well-characterized uniform motion of a rigid sample, as found in various industrial scenarios, where motion is induced by conveyor belts and robotic manipulators. The concept can be easily extended to work with any kind of linear motion and even arbitrary sample trajectories in six degrees of freedom. The applicability to the measurement of morphing surfaces for accurate, point-wise characterized surface deformation is conceivable, but not yet evaluated. For rigid bodies, the optimization may be extended by simultaneous motion estimation, which will be the scope of future work.

CRediT authorship contribution statement

Stefan Hager: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization; **Georg Schitter:** Writing – review & editing, Supervision, Funding acquisition; **Ernst Csencsics:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Stefan Hager reports financial support was provided by Christian Doppler Research Association. Stefan Hager reports financial support was provided by Micro-Epsilon Messtechnik GmbH & Co KG. Stefan Hager reports financial support was provided by Austrian Federal Ministry of Economy, Energy and Tourism. Stefan Hager reports financial support was provided by Austrian National Foundation for Research Technology and Development. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The financial support by the Austrian Federal Ministry of Economy, Energy and Tourism and the National Foundation for Research, Technology and Development and the Christian Doppler Research Association, and Micro-Epsilon Atensor GmbH and MICRO-EPSILON-MESSTECHNIK GmbH & Co.K.G. is gratefully acknowledged.

Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.optlaseng.2026.109627.

References

[1] Sansoni G, Trebeschi M, Docchio F. State-of-the-art and applications of 3D imaging sensors in industry, cultural heritage, medicine, and criminal investigation. Sensors 2009;9(1):568–601.
[2] Brown GM. Overview of three-dimensional shape measurement using optical methods. Opt Eng 2000;39(1):10.
[3] Fan J, Wang X, Zhou C, Ou Y, Jing F, Hou Z. Development, calibration, and image processing of underwater structured light vision system: a survey. IEEE Trans Instrum Meas 2023;72:1–18.
[4] Marrugo AG, Gao F, Zhang S. State-of-the-art active optical techniques for three-dimensional surface metrology: a review. J Opt Soc Am A 2020;37(9):B60.
[5] Chen S. Intraoral 3-D measurement by means of group coding combined with consistent enhancement for fringe projection pattern. IEEE Trans Instrum Meas 2022;71:1–12.
[6] Deng L, Liu G, Zhang Y. A review of machine vision applications in aerospace manufacturing quality inspection. In: 2024 4th International conference on computer, control and robotics (ICCCR). Shnaghai, China: IEEE; 2024, p. 31–9.
[7] Azamfirei V, Psarommatis F, Lagrosen Y. Application of automation for in-line quality inspection, a zero-defect manufacturing approach. J Manuf Syst 2023;67:1–22.
[8] Han L, Li Z, Zhong K, Cheng X, Luo H, Liu G, et al. Vibration detection and motion compensation for multi-frequency phase-shifting-based 3D sensors. Sensors 2019;19(6):1368.
[9] Van Der Jeught S, Dirckx J JJ. Real-time structured light profilometry: a review. Opt Lasers Eng 2016;87:18–31.
[10] Wang Z. Review of real-time three-dimensional shape measurement techniques. Measurement 2020;156:107624.
[11] Salvi J, Pagès J, Batlle J. Pattern codification strategies in structured light systems. Pattern Recognit 2004;37(4):827–49.
[12] Salvi J, Fernandez S, Pribanic T, Llado X. A state of the art in structured light patterns for surface profilometry. Pattern Recognit 2010;43(8):2666–80.
[13] Lavoie P, Ionescu D, Petriu EM. 3-D object model recovery from 2-D images using structured light. IEEE Trans Instrum Meas 2004;53(2):437–43.
[14] Tornero-Martínez N, Trujillo-Schiaffino G, Anguiano-Morales M, Mendoza-Villegas PG, Salas-Peimbert DP, Corral-Martínez LF. Color profilometry techniques: a review. Opt Pura y Aplicada 2018;51(4):1–26.
[15] Morano RA, Ozturk C, Conn R, Dubin S, Zietz S, Nissano J. Structured light using pseudorandom codes. IEEE Trans Pattern Anal Mach Intell 1998;20(3):322–7.
[16] Pagès J, Salvi J, Collewet C, Forest J. Optimised De Bruijn patterns for one-shot shape acquisition. Image Vis Comput 2005;23(8):707–20.
[17] Gu F, Song Z, Zhao Z. Single-shot structured light sensor for 3D dense and dynamic reconstruction. Sensors 2020;20(4):1094.
[18] Li Y, Wang Z. 3D Reconstruction with single-shot structured light RGB line pattern. Sensors 2021;21(14):4819.
[19] Zhang S. Absolute phase retrieval methods for digital fringe projection profilometry: a review. Opt Lasers Eng 2018;107:28–37.
[20] Hansen K, Pedersen J, Solund T, Aanaes H, Kraft D. A structured light scanner for hyper flexible industrial automation. In: 2nd International conference on 3D vision. Tokyo: IEEE; 2014, p. 401–8.
[21] Cheng T, Qin L, Li Y, Hou J, Xiao C. An adaptive multiexposure scheme for the structured light profilometry of highly reflective surfaces using complementary binary gray code. IEEE Trans Instrum Meas 2024;73:1–12.
[22] Catalucci S, Thompson A, Piano S, Branson DT, Leach R. Optical metrology for digital manufacturing: a review. Int J Adv Manuf Technol 2022;120(7–8):4271–90.
[23] Imkamp D, Berthold J, Heizmann M, Kniel K, Peterek M, Schmitt R, et al. Herausforderungen und trends in der Fertigungsmesstechnik - industrie 4.0. tm - Technisches Messen 2016;83(7–8):417–29.
[24] Auerswald MM, Von Freyberg A, Fischer A. Laser line triangulation for fast 3D measurements on large gears. Int J Adv Manuf Technol 2019;100(9–12):2423–33.
[25] Lu L, Suresh V, Zheng Y, Wang Y, Xi J, Li B. Motion induced error reduction methods for phase shifting profilometry: a review. Opt Lasers Eng 2021;141:106573.
[26] Sundar V, Ma S, Sankaranarayanan AC, Gupta M. Single-photon structured light. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). 2022;17865–75.
[27] Zhong K, Li Z, Shi Y, Wang C, Lei Y. Fast phase measurement profilometry for arbitrary shape objects without phase unwrapping. Opt Lasers Eng 2013;51(11):1213–22.
[28] Wang Y, Liu K, Hao Q, Wang X, Lau DL, Hassebrook LG. Robust active stereo vision using Kullback-Leibler divergence. IEEE Trans Pattern Anal Mach Intell 2012;34(3):548–63.
[29] Weise T, Leibe B, Van Gool L. Fast 3D scanning with automatic motion compensation. In: 2007 IEEE Conference on computer vision and pattern recognition. Minneapolis, MN, USA: IEEE; 2007, p. 1–8.
[30] Feng S, Zuo C, Tao T, Hu Y, Zhang M, Chen Q, et al. Robust dynamic 3-D measurements with motion-compensated phase-shifting profilometry. Opt Lasers Eng 2018;103:127–38.
[31] Kayaba H, Kokumai Y. Non-contact full field vibration measurement based on phase-shifting. In: 2017 IEEE Conference on computer vision and pattern recognition (CVPR). Honolulu, HI: IEEE; 2017, p. 2548–56.
[32] Jeon S, Geon Lee H, Sung Lee J, Min Kang B, Wook Jeon B, Young Yoon J, et al. Motion-induced error reduction for motorized digital fringe projection system. IEEE Trans Instrum Meas 2024;73:1–13.
[33] Gumhold S, König S. Image-based motion compensation for structured light scanning of dynamic surfaces. In: Visual Computing - Convergence of Computer Graphics and Computer Vision. Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik; 2008, p. 1–4.
[34] He Q, Ning J, Liu X, Li Q. Phase-shifting profilometry for 3D shape measurement of moving objects on production lines. Precis Eng 2025;92:30–8.
[35] Meza J, Vargas R, Romero LA, Zhang S, Marrugo AG. What is the best triangulation approach for a structured light system? In: Zhang S, Harding KG, editors. Dimensional optical metrology and inspection for practical applications IX. Online Only, United States: SPIE; 2020, p. 12.
[36] Geng J. Structured-light 3D surface imaging: a tutorial. Adv Opt Photonics 2011;3(2):128.
[37] Sansoni G, Carocci M, Rodella R. Three-dimensional vision based on a combination of gray-code and phase-shift light projection: analysis and compensation of the systematic errors. Appl Opt 1999;38(31):6565.
[38] Hartley R, Zisserman A. Multiple view geometry in computer vision. 2nd ed. Cambridge, UK ; New York: Cambridge University Press;; 2003.
[39] Liu Y, Pears N, Rosin PL, Huber P, editors. 3D Imaging, analysis and applications. Cham: Springer International Publishing; 2020.
[40] Gao X, Zhang T. Introduction to visual SLAM: from theory to practice. Singapore: Springer Singapore; 2021.
[41] Feng S, Zuo C, Zhang L, Tao T, Hu Y, Yin W, et al. Calibration of fringe projection profilometry: a comparative review. Opt Lasers Eng 2021;143:106622.
[42] The MathWorks, Inc.. MATLAB optimization toolbox (R2022b). The MathWorks, Inc.; Natick, Massachusetts; 2022. https://www.mathworks.com/products/optimization.html
[43] Lilienblum E, Michaelis B. Optical 3D surface reconstruction by a multi-period phase shift method. J Comput 2007;2(2):73–83.
[44] Zhang Q, Su X, Xiang L, Sun X. 3-D shape measurement based on complementary gray-code light. Opt Lasers Eng 2012;50(4):574–9. Computational Optical Measurement.
[45] Segal A, Haehnel D, Thrun S. Generalized-ICP. In: Robotics: science and systems V. Robotics: Science and Systems Foundation; 2009,.
[46] Andreassen T. Fast_point2trimesh. https://github.com/thor-andreassen/Fast_Point2TriMesh/releases/tag/v1.0.0; 2023. Retrieved January 9, 2025.
[47] An Y, Hyun J-S, Zhang S. Pixel-wise absolute phase unwrapping using geometric constraints of structured light system. Opt Express 2016;24(16):18445–59.